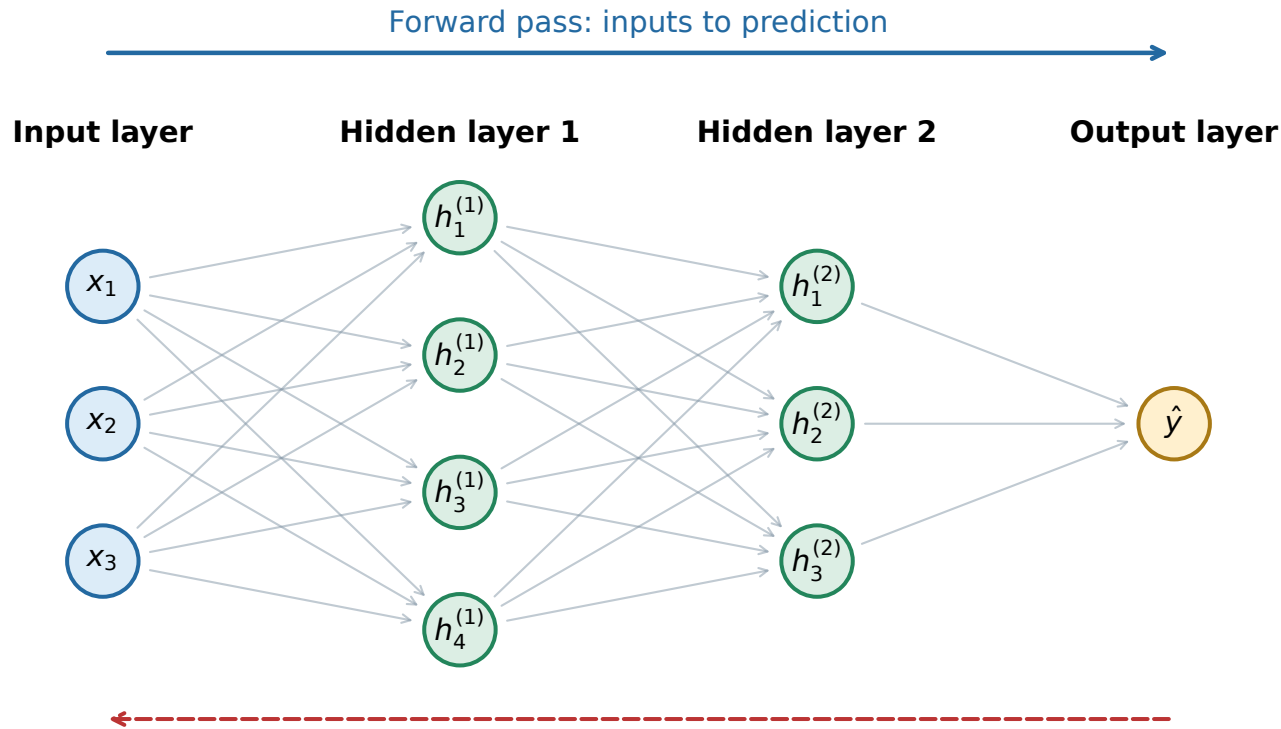

Hidden neuron: $h_j^{(\ell+1)} = \sigma\left(\sum_i W_{ji}^{(\ell)} h_i^{(\ell)} + b_j^{(\ell)}\right)$

Connections carry learned weights; each hidden neuron adds a bias and applies an activation.